

Path Planning and Platooning Control of Connected Automated Vehicles on Curved Roads

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Abstract—This paper is concerned with the safe platooning of multiple connected automated vehicles on curved roads, addressing key challenges such as obstacle and collision avoidance, nonholonomic vehicle dynamics, and compensation for unknown time-varying flowfields. To achieve this, a joint path planning and platooning control approach is proposed to ensure smooth collision-free maneuvers along planned paths while adhering to safety constraints and overcoming the aforementioned challenges. First, a novel scaling-based path planning algorithm is introduced, leveraging obstacle information within an instantiated Gaussian model to re-plan safe paths and define robust safe regions, blue with seamless integration into the vehicle control system. Second, an adaptive nominal controller is developed to achieve precise longitudinal and lateral control objectives. Third, a safety-critical controller is formulated by incorporating control barrier functions into a minimally invasive quadratic programming framework. Finally, numerical experiments are conducted to validate the efficacy of the proposed approach.

Index Terms—Platooning control, path planning, connected automated vehicles, obstacle avoidance, collision avoidance.

I. INTRODUCTION

The rapid advancement of connected automated vehicles (CAVs) is revolutionizing the future of transportation. A key application of CAV technology is platooning, which ensures that a group of vehicles can travel efficiently and closely together while maintaining desired yet safe inter-vehicle distances. CAV platoons can generally enhance traffic efficiency, reduce fuel consumption, and improve road safety by enabling coordinated, closely-spaced travel through automated control and wireless communication.

Most existing platooning control studies of CAVs have primarily focused on cooperative longitudinal control that regulates longitudinal gaps, maintains safe headways, and achieves string stability within the platoon [1]–[5]. This aspect is relatively straightforward as it involves speed and acceleration adjustments to ensure coordinated motion. However, real-world roads have complex geometries, including curves, slopes, and varying lane widths. Lateral control is further

needed to address more diverse and intricate tasks, such as lane changing [6], [7], path planning and following [8], [9]. These tasks are essential for maintaining vehicle stability and alignment with road geometries, especially in dynamic environments with potential obstacles. In this study, the focus is on path planning and following, emphasizing the combined longitudinal and lateral control necessary for safe platooning.

Road obstacles pose a significant challenge to the design of vehicle platooning control strategies and algorithms, driven by the critical safety requirements of autonomous vehicle operations [10]. In global path planning with static obstacles, the A* algorithm [11] and optimization methods like genetic algorithm [12] and differential evolution [13] are devoted to determining optimal paths efficiently, which, however, incurs large computational memory costs. For local path planning, the line-of-sight approach [14], originally developed for missile guidance, produces rigid trajectories between waypoints but has limited flexibility. On the other hand, the potential-field-based obstacle avoidance method [15] is easily prone to local minima, as noted in [16]. Hybrid schemes, such as those in [17], may deviate the robot/vehicle from the desired path. Recently, several learning-based methods, such as Neural RRT* [18] and deep reinforcement learning for dynamic obstacle avoidance [19], exhibit strong capabilities in path planning by leveraging large-scale network training to learn optimal paths. Despite substantial progress, there remains a pressing need for rapid and reliable path planning mechanisms tailored to safe vehicle platooning scenarios, where real-time responsiveness and control-compatible implementation are paramount.

To mitigate collision risks, control barrier function (CBF) has emerged as an effective tool. Safety requirements are typically formulated as constraints within quadratic programming (QP) problems. The CBF-QP framework has been successfully applied to tackle consensus [20] and formation [21] control problems. In [22], a collision avoidance controller was developed by modifying the nominal controller when a collision is truly imminent. This minimally invasive approach was later extended by [23] for multiple robots. To enhance the solvability of QPs, Bayesian optimization was employed in formation control by [24]. Additionally, high-order CBFs were introduced in [25] to extend the method to high relative degree systems. Further advancements in improving the feasibility of optimization problems can be found in [25], [26]. The safety of traffic systems in longitudinal lanes has been addressed in centralized [27] and bidirectional [28] communication settings, of which in lateral direction has been guaranteed by enforcing a minimum deviation from the straight lane in [29]. To the best of the authors' knowledge, the safety of vehicle platooning

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control, particular in the presence of curved path constraints, has not been fully investigated.

Flowfields present another significant challenge to the stability and tracking performance of vehicle control systems. These flowfields can arise from a variety of sources, such as aerodynamic forces, crosswinds, turbulence generated by nearby vehicles, or dynamic airflow changes caused by road features and weather conditions. As noted in [30], these flowfields compromise the accuracy of both path tracking and platoon operations, underscoring the importance of mitigating their impact. To reject the spatially varying flowfields, approximation-based methods such as neural network and fuzzy logic have been widely employed [31], [32]. In [33], the estimation of parameterizable flowfields was studied. For more general time-varying case, [34] introduced an adaptive cooperative control scheme, where the flow parameters were estimated using the congelation of variables method [35]. However, these approaches are mainly applicable to vehicle control systems of integrator-type dynamics. Note that the nonholonomic dynamics present additional complexity, as they require simultaneous achievement of both position and orientation objectives. A consensus control strategy was proposed in [36] for wheeled robots navigating along a planar path, and extensions to parametric paths with aperiodic sampling and actuator faults were made in [37], [38]. Nevertheless, the effects of flowfields are not considered.

In this paper, we aim to develop a safe platooning control approach for multiple CAVs of nonholonomic dynamics with simultaneous safety enhancements including obstacle avoidance, collision avoidance, and flowfield compensation. This ensures that CAVs maintain smooth and collision-free maneuvers along planned paths while adhering to safety constraints and handling the challenges of curved roads and nonholonomic dynamics. The main contribution lies in the *coherent design and analysis* of a scaling-based path planner and a platooning controller under a unified geometric framework. Specifically, 1) *A new scaling-based path planning mechanism* is proposed to guarantee obstacle avoidance. Specifically, the instantiated Gaussian model is utilized to construct a smooth, safe path around multiple obstacles, and a safe region for robustness. This planner is then incorporated into the subsequent vehicle control system synthesis. 2) *An adaptive control design* is proposed for the nominal longitudinal and lateral control of non-holonomic vehicles operating in unknown, time-varying parametric flowfields. A formal stability analysis is also conducted, demonstrating the asymptotic convergence of the longitudinal and lateral control systems. 3) *A safety-guaranteed CBF-QP control scheme* is developed, where CBFs are designed to formulate the safety constraints, ensuring minimal interference with the nominal controller while effectively addressing both control objectives and safety requirements.

II. PRELIMINARIES

A. Vehicle-to-Vehicle Communication Topologies

We consider a CAV platoon consisting of one virtual vehicle, known as the leader, and N following vehicles, known as the followers. Our primary control objective is to design a

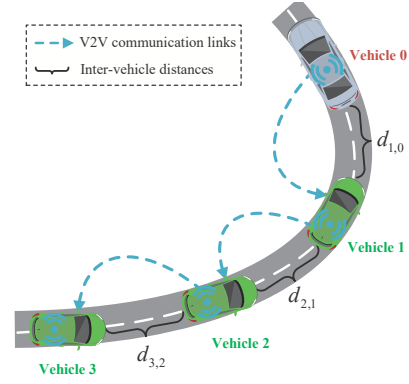


Fig. 1. An illustration of vehicle platooning on a curved road.

path planning and platooning control approach such that each follower can be regulated to track the leader while keeping the desired safe inter-vehicle distance from their predecessors. The platoon will be coordinated to travel along a pre-assigned path that will be specified later. Fig. 1 illustrates such a platoon scenario involving four CAVs on a curved road.

Information sharing among CAVs plays an important role in accomplishing the above control objective. We employ a digraph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{A}\}$ with $N + 1$ nodes to describe the vehicle-to-vehicle (V2V) communication topology. Here, $\mathcal{V} = \{0, 1, \dots, N\}$ denotes the node set, $\mathcal{E} = \mathcal{V} \times \mathcal{V}$ represents the edge set, and $\mathcal{A} = [a_{ij}]_{i,j=0}^N$ signifies the adjacency matrix. The element $a_{ij} = 1$ if $(i, j) \in \mathcal{E}$, which indicates the information flow from node j to node i , and $a_{ij} = 0$, otherwise. Let $\mathcal{N}_i$ be the neighborhood set of node i . Define the Laplacian matrix as $\mathcal{L} = [l_{ij}]_{i,j=1}^N$, where $l_{ij} = -a_{ij}$ and $l_{ii} = \sum_{j \in \mathcal{N}_i} a_{ij}$. Furthermore, define a matrix $\mathcal{H} = \mathcal{L} + \mathcal{B}$, with $\mathcal{B} = \text{diag}\{a_{10}, \dots, a_{N0}\}$. A link is a sequence of edges $(i, l), \dots, (k, j)$ with distinct nodes. A digraph has a directed spanning tree, if there exists a root node that has directed links to all other nodes.

Assumption 1: The digraph $\mathcal{G}$ comprises a directed spanning tree, with the leader vehicle serving as the root.

B. Longitudinal and Lateral Vehicle Dynamics

We assume that each CAV has unicycle-type dynamics, which can be represented by the following equations [39]:

$$\begin{cases} \dot{x}_i(t) = v_i(t) \cos \theta_i(t) \\ \dot{y}_i(t) = v_i(t) \sin \theta_i(t) \\ \dot{\theta}_i(t) = \omega_i(t) + f_{\theta_i}(t) \\ \dot{v}_i(t) = a_i(t) \end{cases} \quad (1)$$

for $i = 0, \dots, N$, where (x_i, y_i) denotes the coordinate of vehicle in the Cartesian coordinate system, $f_{\theta_i} : \mathbb{R}^2 \rightarrow \mathbb{R}$ is an unknown time-varying rotating flowfield to be specified, θ_i represents the angle between the longitudinal velocity v_i and positive x-axis, and $a_i \in \mathbb{R}$ and $\omega_i \in \mathbb{R}$ are acceleration and angular velocity inputs to be determined, satisfying

$$a_{\min} \leq a_i \leq a_{\max}, \quad \omega_{\min} \leq \omega_i \leq \omega_{\max} \quad (2)$$

with $a_{\min}, a_{\max}, \omega_{\min}, \omega_{\max}$ being known constants.

The Frenet coordinate system is a powerful and widely used approach to transform the distances along the curve and

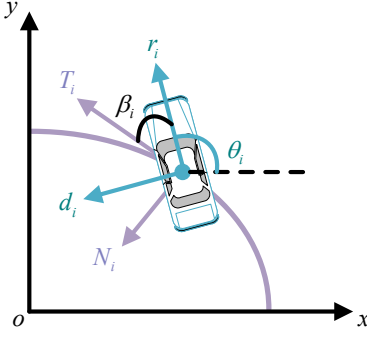


Fig. 2. Coordinate transformation in the Frenet coordinate system.

perpendicular to the curve. Let $p_i = [x_i, y_i]^T \in \mathbb{R}^2$, and define $r_i = [\cos \theta_i, \sin \theta_i]^T \in \mathbb{R}^2$ and $d_i = [-\sin \theta_i, \cos \theta_i]^T \in \mathbb{R}^2$ as the tangent and normal vectors of the current movement. Then the dynamics of vehicle (1) can be written in the Frenet coordinate system as

$$\begin{cases} \dot{p}_i(t) = v_i(t)r_i(t) \\ \dot{r}_i(t) = (\omega_i(t) + f_{\theta_i}(t))d_i(t) \\ \dot{d}_i(t) = -(\omega_i(t) + f_{\theta_i}(t))r_i(t) \\ \dot{v}_i(t) = a_i(t). \end{cases} \quad (3)$$

The coordinate transformation is shown in Fig. 2, where N_i and T_i are the normal vector and tangent vector to each curve, respectively, and $\beta_i \in (-\pi, \pi]$ is the angle between r_i and T_i . One has $\cos \beta_i = r_i \cdot T_i = d_i \cdot N_i$ and $\sin \beta_i = d_i \cdot T_i = -r_i \cdot N_i$.

C. High-Order Control Barrier Function

Consider an affine control system described by

$$\dot{x} = f(x) + g(x)u, \quad (4)$$

where $x \in \mathcal{U}_x \in \mathbb{R}^n$, $u \in \mathcal{U}_u \in \mathbb{R}^q$, and $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $g : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times q}$ are globally Lipschitz.

For a differentiable function $b : \mathbb{R}^n \rightarrow \mathbb{R}$ with respect to the constraint $b(x)$ of relative degree $m \geq 1$ for system (4), we define a sequence function $\psi_k : \mathbb{R}^n \rightarrow \mathbb{R}$ as

$$\psi_k(x) = \dot{\psi}_{k-1}(x) + \alpha_k(\psi_{k-1}(x)) \quad (5)$$

for $k = 1, \dots, m$ with $\psi_0(x) = b(x)$, where $\alpha_k(\cdot)$ denotes a class $\mathcal{K}$ function satisfying $\alpha_k(r) < r$ for all $r > 0$. Subsequently, a sequence of sets associated with (5) can be constructed as

$$\mathcal{S}_k = \{x \in \mathbb{R}^n \mid \psi_{k-1}(x) \geq 0\}. \quad (6)$$

Definition 1: Let $\mathcal{S}_k$ be defined by (6) and $\psi_k(x)$ be defined by (5) for $k \in \{1, \dots, m\}$. A function $b : \mathbb{R}^n \rightarrow \mathbb{R}$ is a *high-order control barrier function (HOCBF)* of relative degree m for system (4), if there exists a differentiable class $\mathcal{K}$ function α_k such that

$$\mathcal{L}_f^m b(x) + \mathcal{L}_g \mathcal{L}_f^{m-1} b(x)u + \mathcal{O}(b(x)) + \alpha_m(\psi_{m-1}(x)) \geq 0 \quad (7)$$

for all $x \in \bigcap_{i=1}^m \mathcal{S}_i$, where $\mathcal{O}(\cdot) = \sum_{i=1}^{m-1} \mathcal{L}_f^i (\alpha_{m-i} \circ \psi_{m-i-1})(x)$, and $\mathcal{L}_g$ and $\mathcal{L}_f$ denote the Lie derivatives along g and f , respectively.

Lemma 1: [25] Given an HOCBF $b : \mathbb{R}^n \rightarrow \mathbb{R}$ of relative degree m with sets $\mathcal{S}_k$, $k \in \{1, \dots, m\}$. For any $x(t_0) \in$

$\bigcap_{i=1}^m \mathcal{S}_i$, the controller $u \in \mathcal{U}_u$ satisfying (7) will render the set $\bigcap_{i=1}^m \mathcal{S}_i$ forward invariant for system (4).

III. PROBLEM FORMULATION

A. Path Description

The desired path followed by the platoon is represented by a simple, closed curve with a consistent, nonzero curvature. In a global coordinate system, such a curve can be parameterized by a smooth mapping, denoted as $\mathcal{C}_0 : \phi \in [0, 2\pi) \rightarrow \mathcal{C}_0(\phi) \in \mathbb{R}^2$. Here, ϕ refers to the *phrase angle* parameter, indicating the direction from the origin to the point on the curve.

By extending $\mathcal{C}_0(\phi)$ along its normal vector, a set of level curves can be derived from $\mathcal{C}_0$ using the concentric compression strategy [34] as:

$$\mathcal{C}_c(\phi, c) = (1 - c)\mathcal{C}_0(\phi), \quad (8)$$

where $c \in \mathbb{R}$ denotes the *compression margin*. Moreover, there exist an open set $\Omega \in \mathbb{R}^2$, which forms a tubular neighborhood around the curve and is constructed as

$$\Omega = \{p \in \mathbb{R}^2 \mid |\lambda(p)| \leq \varepsilon\} \quad (9)$$

and a smooth function (also called the *curve function*) $\lambda : \Omega \rightarrow (-\varepsilon, \varepsilon)$ such that $\lambda(p) = c$ denotes all points $p = (x, y)$ on the curve identified by $\mathcal{C}_c$. Additionally, the *arc-length* $s(\lambda, \phi)$ along the curve specifies the position of the vehicle on the path, which is defined by

$$s(\lambda, \phi) \triangleq \int_{\phi^*}^{\phi} \frac{\partial s(\lambda, \tau)}{\partial \tau} d\tau, \quad (10)$$

where ϕ^* corresponds to the parameter associated with the starting point. The relationship of the aforementioned notations is illustrated in Fig. 3.

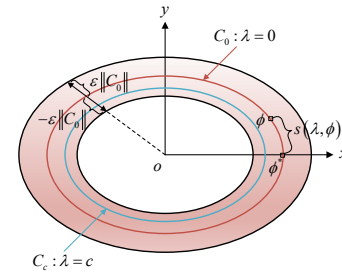


Fig. 3. An illustration of the level curves of $\mathcal{C}_0$ on its tubular neighborhood Ω , where the red gradient area denotes the Ω .

B. Flowfield Parameterization

In this paper, we consider that the time-varying flowfield $f_{\theta_i}(t)$ in (1) can be parameterized as

$$f_{\theta_i}(t) = f_{a_i}(\lambda_i(t), \beta_i(t)) f_{b_i}(p_i(t), t) \vartheta_i(t), \quad (11)$$

where $f_{a_i} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ denotes the known flow amplitude and $f_{b_i} : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}^q$ represents the known basis function of the flowfield. Here, $\vartheta_i(t) \in \mathbb{R}^q$ is an unknown time-varying parameter vector without any restrictions on its variation, and belongs to a compact set Θ_i , the radius of which is δ_{θ_i} , naturally unknown.

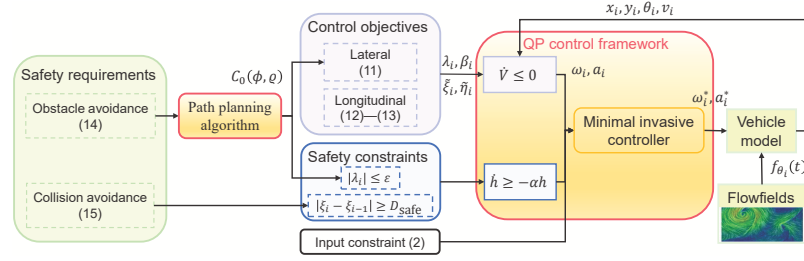


Fig. 4. A schematic of the path planning and platooning control approach.

Assumption 2: The flow amplitude f_{a_i} is bounded and satisfies $\beta_i \cdot f_{a_i}(\cdot, \beta_i) \geq 0$ for all $t \geq 0$, and $f_{a_i}(0, 0) = 0$.

Assumption 3: There exists a constant $c_0 > 0$ such that the flow basis $\|f_{b_i}(\cdot, \cdot)\| \leq c_0$ for all $t \geq 0$.

Remark 1: Assumption 2 is mild as one can choose f_{a_i} to guarantee that the sign is consistent with β_i . On the other hand, $f_{a_i}(0, 0) = 0$ implies that the effect of the flowfield disappears when the vehicle moves along the tangent direction of the desired path. Assumption 3 represents a commonly-adopted boundedness condition [33], [34].

Remark 2: The dependence of the flowfield model (11) on the lateral deviation λ_i and heading angle β_i is without loss of generality [40]. Specifically, crosswinds or airflow disturbances near the roadside, caused by guardrails, terrain features, or nearby vehicles, can induce lateral forces that vary with the vehicle's distance from the road centerline. On the other hand, misalignment between the vehicle heading and road direction may change the tire forces and slip angles, thereby affecting the aerodynamic forces and slip-induced lateral dynamics. Moreover, this model can be used to describe the spatially varying flowfield in [41], the time-invariant parametric flowfield in [33].

C. Control Objectives with Safety Requirements

The control objectives of this study include both the longitudinal and lateral aspects, while ensuring safety requirements in terms of obstacle and collision avoidance.

Control objectives: Lateral control is achieved if along the trajectories of (3) the following conditions hold:

$$\lim_{t \rightarrow \infty} \lambda_i(t) = 0, \quad \lim_{t \rightarrow \infty} \beta_i(t) = 0. \quad (12)$$

In line with [34], [42], the *generalized arc-lengths* $\xi_i(s_i)$ are defined as $\mathbb{C}^2$ functions of s_i to formulate the longitudinal control into a consensus problem, where $|\partial \xi_i / \partial s_i| > 0$ and $\partial^2 \xi_i / \partial s_i^2$ is uniformly bounded. Longitudinal control is achieved if

$$\lim_{t \rightarrow \infty} (\xi_i(s_i(t)) - \xi_{i-1}(s_{i-1}(t))) = d_{i,i-1}(t), \quad (13)$$

$$\lim_{t \rightarrow \infty} (\eta_i(t) - \eta_*(t)) = 0, \quad (14)$$

where $d_{i,i-1}(t)$ represents the desired distance between two CAVs i and $i-1$, η_i denotes the derivative of ξ_i along the path, and $\eta_*(t)$ is the reference speed of the platoon, with a bounded derivative $\tau_*(t) = \dot{\eta}_*(t)$. In this study, a constant spacing is considered, which yields $d_{i,i-1}(t) \equiv d_0 > 0$.

Safety requirements: Obstacle avoidance is ensured by maintaining a minimum distance between each CAV and any

obstacle, while collision is prevented by maintaining a safe longitudinal spacing between CAVs along the desired path. These are formulated as

$$\|p_i(t) - p_{\text{ob},j}(t)\| > R_{\text{ob},j} > 0, \quad j = 1, \dots, M, \quad (15)$$

$$|\xi_i(s_i(t)) - \xi_{i-1}(s_{i-1}(t))| > D_{\text{safe}} > 0, \quad (16)$$

where $p_{\text{ob},j}$ denotes the j -th obstacle's position, M is the number of obstacles, $R_{\text{ob},j}$ and D_{safe} are positive constants.

The main problem to be addressed can now be stated below.

Problem 1: Suppose that Assumptions 1–3 hold. For any given path, determine a feasible platooning control policy (a_i, ω_i) for CAVs described by (1) such that under the input constraint (2) and the flowfields (11), the control objectives (12)–(14) and the safety requirements (15)–(16) are satisfied.

IV. MAIN RESULTS

In this section, we first present a new path planning algorithm for successful obstacle avoidance. Next, we provide the detailed design of the nominal combined longitudinal and lateral controllers that guarantee the closed-loop system stability but without safety guarantees for the platoon. Finally, we develop a safety-guaranteed platooning controller based on the CBF-QP optimization, as illustrated in Fig. 4.

A. A New Scaling-Based Path Planning Algorithm

In a mixed traffic environment, the CAV platoon on the road must dynamically adjust its paths to avoid obstacles, minimize risks with other road users, and enhance fuel efficiency. Our path planning algorithm includes re-planning a safe path via the use of a path scaling technique, effectively “excluding” obstacles from the original path, as illustrated in Fig. 5. The path scaling technique generates various paths for the platoon by introducing a smooth scaling factor $\varrho(t) \in \mathbb{R}$. By incorporating this scaling factor into the parametric representation of the original path $C_0(\phi)$, we obtain the corresponding scaled path, described by the smooth mapping:

$$C'_0 : \phi \in [0, 2\pi) \times \varrho \in \mathbb{R} \rightarrow C'_0(\phi, \varrho) \in \mathbb{R}^2, \quad (17)$$

where $\varrho(t) \in \mathbb{R}$ can be specified according to user requirements. Naturally, the tubular neighborhood Ω and the curve function λ associated with $C'_0(\phi, \varrho)$ follow the same definitions as the original path. Fig. 6 provides an example of the level superellipses (23) with different selections of ϱ , and Algorithm 1 states how to select ϱ for safe path planning.

Note that the state-based Gaussian model represents a key component in the proposed algorithm. This is because the center $c_{\varrho,j}$ and width $w_{\varrho,j}$ parameters of the model align well with

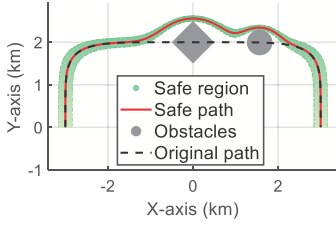


Fig. 5. An illustration of the proposed path planning mechanism for obstacle avoidance.

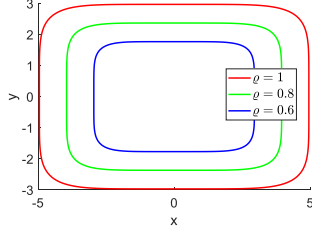


Fig. 6. An illustration of level superellipses with different scaling factors under $C'_0(\phi, \rho)$ in (23), where $a = 5$, $b = 3$, $\mu = 0$, and $n = 3$.

the available obstacle information, simplifying the parameter tuning process. Furthermore, the smoothness of the instantiated Gaussian model ensures that the path remains smooth, thus the designed controller is prevented from chattering due to discontinuities. Hence, Algorithm 1 generates a safe path $C'_0(\phi, \rho)$ that avoids all obstacles, with a corresponding safe region Ω constructed around it. Considering the constraints of road boundaries, the bound ε should be chosen to be no larger than half of the lane width, such that the tubular neighborhood Ω is a subset of the drivable lane. Then the scaled path and its safe region remain strictly inside the road boundaries.

Although changing lanes is a natural strategy for avoiding obstacles in multi-lane roads, adjacent lanes may not exist or be accessible in certain scenarios. The proposed path-planning algorithm is not intended to replace lane-change decision making. Instead, it provides a continuous and smooth deformation of the current reference path so that the platoon stays close to the nominal path while avoiding obstacles. Under Algorithm 1, the CAVs can be regulated simultaneously along the safe path $C'_0(\phi, \rho)$ and within designated safe region Ω , which ensures the obstacle avoidance requirement (15). Compared to directly imposing position constraint (15) on the controller, the proposed algorithm offers at least two key advantages. First, the obstacle avoidance requirement is seamlessly integrated into the lateral control objectives defined in (12), ensuring that the original nominal control framework remains intact. Second, the algorithm features a robustly designed safe region that accounts for flowfields and disturbances, while offering the flexibility to deactivate this mechanism when these factors are absent. While the safe region enforces the position constraint $p_i \in \Omega$ on the CAVs, it can be equivalently reformulated as an output constraint, i.e.,

$$p_i \in \Omega \Leftrightarrow |\lambda_i| \leq \varepsilon, \quad (22)$$

thus significantly improving the feasibility. The detailed implementation of this approach is elaborated in Section IV-C.

Remark 3: Consider M obstacles and a path discretized into N_p evaluation points. Let K denote the number of iterations

Algorithm 1: Scaling-based path planning

1 **Input:** unsafe original path $C_0(\phi)$.
2 **Output:** safe path $C'_0(\phi, \rho)$.
3 **Initialize:** number of obstacles M , obstacle positions $p_{ob,j}$, obstacle radii $R_{ob,j}$, Gaussian model with amplitude $A_{e,j} > 0$, center $c_{e,j} = p_{ob,j}$, width $w_{e,j} = R_{ob,j}$, safe margin ε , learning rate $0 < \eta_A < 1$, coefficient $\alpha_A > 0$, tolerance $\delta > 0$, and $convergence_id = 0$.
4 **while** $convergence_id == 0$ **do**
5 Select the scaling factor using the Gaussian model as

$$\varrho(p) = 1 + \sum_{j=1}^M A_{e,j} \exp \left[-\frac{\|p - c_{e,j}\|^2}{2w_{e,j}^2} \right]. \quad (18)$$

6 Construct the path $C'_0(\phi, \rho)$ and its corresponding curve function $\lambda(p, \rho)$ by (18). // Remark 6 provides a specific example of C'_0 and λ
7 Update the parameters $A_{e,j}$ based on gradient descent

$$A_{e,j} \leftarrow A_{e,j} + \eta_A \frac{\partial J}{\partial \rho} \frac{\partial \rho}{\partial A_{e,j}} \quad (19)$$

by minimizing the cost function

$$J = \sum_{j=1}^M \left[(\|p - p_{ob,j}\| - R_{ob,j} - \varepsilon)^2 + \alpha_A A_{e,j}^2 \right]$$

s.t. $\lambda(p, \rho) = 0$. (20)
8 Compute the minimum distance from the obstacle to the path as

$$D = \min_{p \in C'_0(\phi, \rho)} \{ \|p - p_{ob,j}\| - R_{ob,j} - \varepsilon \}, \forall j. \quad (21)$$

9 **if** $D \geq -\delta$ **then**
10 | $convergence_id = 1$
11 **end**
12 **end**

until convergence. In each iteration, the path deformation is applied point-wise, and the scaling factor $\rho(p)$ involves M Gaussian terms, yielding a cost of $\mathcal{O}(M \cdot N_p)$. The gradient update for each Gaussian amplitude parameter requires $\mathcal{O}(M)$, while the evaluation of the convergence condition costs $\mathcal{O}(M \cdot N_p)$. Altogether, the total complexity of Algorithm 1 is $\mathcal{O}(K \cdot M \cdot N_p)$. In contrast, the A* algorithm has a complexity of $\mathcal{O}(E + V \log V)$, while the RRT algorithm requires an average cost of $\mathcal{O}(n \log n)$, where E and V denote the numbers of vertices and edges in the grids, respectively, and n is the number of sampled nodes. Therefore, the proposed path planning algorithm exhibits lower computational complexity for large map size and high resolution scenarios.

Remark 4: The proposed path planning algorithm for curved road scenarios differs from some existing algorithms in the following aspects. 1) Seamless integration with control systems: The planned path is generated through continuous scaling and deformation of a nominal reference path, maintaining smoothness and differentiability. This allows the path to be directly embedded into the path-related controller without requiring additional post-processing. This sets it apart from the grid-based methods (e.g., A* [11]) and sampling-based (e.g., RRT [18]) often produce piecewise or discrete paths that necessitate interpolation or re-smoothing for use in control applications. 2) Flexibility in geometric design: The proposed algorithm adopts a flexible curve that can be extended in both longitudinal and lateral directions. This flexibility contrasts with strictly longitudinal paths considered in previous studies, such as [43],

[44], offering greater adaptability to varying road geometries and obstacle configurations. 3) Low computational complexity: The planned path is explicitly parameterized by a spatial scaling factor, and the optimization process involves only one parameter, $A_{e,j}$, for obstacle avoidance adjustment. This concise formulation enhances interpretability, avoids the risk of local minima in potential field methods [15], and reduces computational burden and tuning effort in most learning-based algorithms [18], [45], which typically require extensive model training and hyperparameter selection. Overall, Algorithm 1 is suitable for efficiently planning safe paths in environments, where the path geometry is known *a priori*, with emphasizing responsiveness and generating smooth, safe, and control-compatible trajectories rather than on global optimality.

Remark 5: The scaling-based path planning algorithm can be extended to handle moving obstacles. Consider the obstacle's position as a time-dependent variable $p_{ob,j}(t)$. The scaling function in equation (18) can be accordingly reformulated as $\rho(p, t) = 1 + \sum_{j=1}^M A_{e,j} \exp \left[-\frac{\|p - c_{e,j}(t)\|^2}{2w_{e,j}^2} \right]$, with the center $c_{e,j}(t)$ chosen as $p_{ob,j}(t)$. This formulation yields a temporal deviation in the path-following error λ_i , and to account for this, an additional compensation term $\frac{\partial \lambda_i}{\partial \rho} \frac{\partial \rho}{\partial t}$ should be incorporated into the nominal controller.

Remark 6: The (skewed) superellipse, also known as the Lamé curve, is a generalized geometric curve ranging from a rectangle to an ellipse. Integrated with the scaling factor ρ , this curve can be described by

$$\begin{aligned} C'_0(\phi, \varrho) &= [x(\phi, \varrho), y(\phi, \varrho)]^T \\ &= \left[a\varrho \cos(\phi)^{\frac{1}{n}} + \mu b\varrho \sin(\phi)^{\frac{1}{n}}, b\varrho \sin(\phi)^{\frac{1}{n}} \right]^T, \end{aligned} \quad (23)$$

where μ denotes the skew factor, n determines the corner sharpness, a and b are the semi-major and semi-minor axes, respectively, with $a \geq b > 0$. Given that the level curve, identified by value c , is obtained by (8), it follows that

$$C'_c(\phi, \varrho, c) := \left(\frac{x - \mu y}{a\varrho} \right)^{2n} + \left(\frac{y}{b\varrho} \right)^{2n} = (1 - c)^{2n}. \quad (24)$$

Therefore, the corresponding curve function becomes

$$\lambda(p, \varrho) = c = 1 - \left(\left(\frac{x - \mu y}{a\varrho} \right)^{2n} + \left(\frac{y}{b\varrho} \right)^{2n} \right)^{\frac{1}{2n}}. \quad (25)$$

B. Nominal Combined Longitudinal and Lateral Controller Design

The derivation of path following error is given by

$$\dot{\lambda}_i = \nabla \lambda_i \cdot \dot{p}_i = \|\nabla \lambda_i\| N_i^T v_i r_i = -\|\nabla \lambda_i\| v_i \sin \beta_i, \quad (26)$$

where $N_i = \frac{\nabla \lambda_i}{\|\nabla \lambda_i\|}$. Taking the time derivative of β_i yields

$$\dot{\beta}_i = -v_i (c_{a,i} \cos \beta_i + c_{b,i} \sin \beta_i) - f_{a,i} f_{b,i} \vartheta_i(t) - \omega_i, \quad (27)$$

where $c_{a,i} = \frac{1}{\|\nabla \lambda_i\|} T_i^T \nabla^2 \lambda_i T_i$, $c_{b,i} = -\frac{1}{\|\nabla \lambda_i\|} T_i^T \nabla^2 \lambda_i N_i$, and $\nabla^2 \lambda_i$ denotes the Hessian matrix of $\lambda_i(p_i)$. Differentiating the generalized arc-length ξ_i along (3) yields

$$\dot{\xi}_i = \frac{\partial \xi_i}{\partial s_i} v_i \left(\cos \beta_i - \frac{\partial s_i}{\partial \lambda_i} \|\nabla \lambda_i\| \sin \beta_i \right). \quad (28)$$

Note that $\beta_i = 0$ if vehicle i moves along its given orbit, and thus $\dot{\xi}_i = \frac{\partial \xi_i}{\partial s_i} v_i$. Then, by defining a variable as $\eta_i = \frac{\partial \xi_i}{\partial s_i} v_i$, the equation (28) can be rewritten as

$$\dot{\xi}_i = \eta_i \left(\cos \beta_i - \frac{\partial s_i}{\partial \lambda_i} \|\nabla \lambda_i\| \sin \beta_i \right). \quad (29)$$

The time derivative of η_i can be calculated by

$$\dot{\eta}_i = \frac{\partial \xi_i}{\partial s_i} a_i + \frac{\partial^2 \xi_i}{\partial s_i^2} v_i^2 \left(\cos \beta_i - \frac{\partial s_i}{\partial \lambda_i} \|\nabla \lambda_i\| \sin \beta_i \right). \quad (30)$$

As a result, the equations associated with the open-loop system of vehicle i are presented in (26), (27), (29), and (30).

It is assumed that the leader vehicle moves along its virtual path for all $t \geq 0$, which serves as a virtual reference to guide the motion of the follower vehicles. Thus, $\lambda_0(p_0) = 0$ and $\beta_0 = 0$. Then the control command for the leader vehicle is designed as

$$a_0 = -\left(\frac{\partial \xi_0}{\partial s_0} \right)^{-1} \left(\frac{\partial^2 \xi_0}{\partial s_0^2} v_0^2 - \tau_* \right), \quad \omega_0 = -v_0 c_{a,0}, \quad (31)$$

which yields the closed-loop system $\dot{\lambda}_0(p_0) = 0$, $\dot{\beta}_0 = 0$, $\dot{\xi}_0 = \eta_0 = \eta_*$, and $\dot{\eta}_0 = \tau_*$.

Note that the reference signal ξ_0 is not accessible to all the follower CAVs. Let $\hat{\xi}_i$ and $\hat{\eta}_i$ be the estimates of ξ_0 and η_0 for each follower CAV i , respectively. Design the distributed estimator as

$$\begin{cases} \dot{\hat{\xi}}_i = \hat{\eta}_i \\ \dot{\hat{\eta}}_i = -\mu_1 \sum_{j \in \mathcal{N}_i} a_{ij} (\hat{\xi}_i - \hat{\xi}_j) - \mu_2 \sum_{j \in \mathcal{N}_i} a_{ij} (\hat{\eta}_i - \hat{\eta}_j) + \tau_*, \end{cases} \quad (32)$$

where μ_1 and μ_2 are positive constants. Denoting by $\tilde{\xi}_i = \hat{\xi}_i - \xi_0$ and $\tilde{\eta}_i = \hat{\eta}_i - \eta_0$ the estimation errors with $\tilde{\xi} = [\tilde{\xi}_1, \dots, \tilde{\xi}_N]^T$ and $\tilde{\eta} = [\tilde{\eta}_1, \dots, \tilde{\eta}_N]^T$, we have that

$$\begin{bmatrix} \dot{\tilde{\xi}} \\ \dot{\tilde{\eta}} \end{bmatrix} = \begin{bmatrix} 0 & I_N \\ -\mu_1 \mathcal{H} & -\mu_2 \mathcal{H} \end{bmatrix} \begin{bmatrix} \tilde{\xi} \\ \tilde{\eta} \end{bmatrix}, \quad (33)$$

yielding that the estimation errors are exponentially convergent, which has been shown in [46, Lemma 2].

Theorem 1: Suppose that Assumptions 1–3 hold. For any initial position $p_i(0) \in \Omega_i$ of (1), the control objectives (12)–(14) are satisfied under the following nominal controller

$$\omega_i = k_1 \sin \frac{\beta_i}{2} - f_{a,i} f_{b,i} \hat{\vartheta}_i + c_0 f_{a,i} \hat{\delta}_i + \Delta_{i,1} \quad (34)$$

$$a_i = -\left(\frac{\partial \xi_i}{\partial s_i} \right)^{-1} [k_2 (\eta_i - \hat{\eta}_i) + \Delta_{i,2}], \quad (35)$$

where k_1 , k_2 are positive gain parameters, Δ_1 and Δ_2 are specified in (41), and the updating laws of adaptive parameters $\hat{\vartheta}_i$ and $\hat{\delta}_i$ are given in (43).

Proof: See Appendix A.

Remark 7: The nominal controller (34) and (35) involves several parameters that influence system behavior. Specifically, the control gains k_1 and k_2 regulate the convergence rate of the heading and speed tracking error, while the adaptation gains γ_1 and γ_2 govern the convergence of the adaptive parameters. Additionally, compensation gain c_1 improves heading

alignment with the intended path, and c_2 contributes to both heading correction and velocity regulation. While increasing these gains can improve responsiveness and tracking accuracy, but overly large values may lead to instability or oscillations.

Remark 8: To enhance the resilience of the platooning controller (34)-(35) against intermittent denial-of-service (DoS) attacks on V2V communication networks, one may employ intelligent *data packet analysers* [2] into the proposed distributed estimators in (32) so as to detect attack occurrence, record data packet arrival times, and store the maximum attack duration. In this case, distributed estimators on CAVs are well informed about the packet arrival times and able to use the trustworthy data packets previously arrived. It is then expected that uniform ultimate boundedness of the platooning tracking control errors can be preserved.

C. Safe Platooning Controller Design

We next establish CBFs for each follower CAV to avoid potential obstacles and collisions with nearby CAVs.

Unifying the dynamics of λ_i in (26) and β_i in (27), we have the following control affine form

$$\dot{\mathbf{x}}_i(t) = \mathbf{f}(\mathbf{x}_i(t)) + \mathbf{g}(\mathbf{x}_i(t))\omega_i, \quad (36)$$

where $\mathbf{x}_i(t) = [\lambda_i, \beta_i]^T$, $\mathbf{f}(\mathbf{x}_i(t)) = [\|\nabla\lambda_i\|v_i \sin\beta_i, -v_i(c_{a,i} \cos\beta_i + c_{b,i} \sin\beta_i) - f_{a,i}f_{b,i}\vartheta_i(t)]^T$, and $\mathbf{g}(\mathbf{x}_i(t)) = [0, -1]^T$. It is straightforward to verify that the relative degree of system (36) is 2, as it needs to be differentiated twice along its dynamics before the controller ω_i explicitly appears in the corresponding derivative. A candidate HOCBF imposed on constraint (15) for system (36) then can be given as

$$b(\mathbf{x}_i) = \varepsilon^2 - \lambda_i^2. \quad (37)$$

The safety sets with respect to (37) are obtained as $S_{1,i} = \{\mathbf{x}_i \in \mathbb{R}^n \mid b(\mathbf{x}_i) \geq 0\}$ and $S_{2,i} = \{\mathbf{x}_i \in \mathbb{R}^n \mid \psi_1(\mathbf{x}_i) \geq 0\}$, with $\psi_1(\mathbf{x}_i) = b(\mathbf{x}_i) + \alpha_1(b(\mathbf{x}_i))$. According to Lemma 1, the CBF condition is given as

$$\mathcal{L}_f^2 b(\mathbf{x}_i) + \mathcal{L}_g \mathcal{L}_f b(\mathbf{x}_i)\omega_i + \mathcal{O}(b(\mathbf{x}_i)) + \alpha_2(\psi_1(\mathbf{x}_i)) \geq 0. \quad (38)$$

Define the relative information between any vehicle pair (i, j) , $j \in \mathcal{N}_i$ as $\delta_{\xi_{ij}} = \xi_i - \xi_j$ and $\delta_{\eta_{ij}} = \eta_i - \eta_j$. When the platoon drives on the desired path, each pair of CAVs (i, j) must maintain a safe distance along the path, denoted as $\|\delta_{\xi_{ij}}\| \geq D_{\text{safe}} > 0$. The vehicle dynamics can be obtained from (29) and (30) as $\dot{\xi}_i = \eta_i$, $\dot{\eta}_i = \tau_i$, where $\tau_i = \frac{\partial \xi_i}{\partial s_i} a_i + \frac{\partial^2 \xi_i}{\partial s_i^2} v_i^2$. It can be readily verified that $\|\tau_i\|_\infty \leq b_\tau$. In this sense, a valid CBF derived from [23] to formulate the safety constraint (16) can be given as $h_{ij}(\xi_i, \eta_i) = \frac{\delta_{\xi_{ij}}}{\|\delta_{\xi_{ij}}\|} \delta_{\eta_{ij}} + \sqrt{2b_\tau(\|\delta_{\xi_{ij}}\| - D_{\text{safe}})}$. To ensure safety, the CBF should satisfy $\dot{h}_{ij}(\cdot) = \frac{1}{\|\delta_{\xi_{ij}}\|} [\delta_{\xi_{ij}} \tau_i + \delta_{\xi_{ij}} \tau_j + \frac{b_\tau \delta_{\xi_{ij}} \delta_{\eta_{ij}}}{\sqrt{2b_\tau(\|\delta_{\xi_{ij}}\| - D_{\text{safe}})}}] \geq -\alpha(h_{ij}(\cdot))$. Based on the findings in [24], the CBF condition for vehicle i can be expressed as

$$-\delta_{\xi_{ij}} a_i \leq \left(\frac{\partial \xi_i}{\partial s_i} \right)^{-1} \left(\frac{\|\delta_{\xi_{ij}}\|}{2} \alpha(h_{ij}(\xi_i, \eta_i)) + \Phi_{ij}^{\text{cbf}} \right), j \in \mathcal{N}_i, \quad (39)$$

$$\text{where } \Phi_{ij}^{\text{cbf}} = \frac{1}{2} \frac{b_\tau \delta_{\xi_{ij}} \delta_{\eta_{ij}}}{\sqrt{2b_\tau(\|\delta_{\xi_{ij}}\| - D_{\text{safe}})}} + \delta_{\xi_{ij}} \frac{\partial^2 \xi_i}{\partial s_i^2} v_i^2.$$

Combining the nominal controller (34) and (35) as well as the CBF conditions, the QP-based controller is designed as:

$$(a_i^*, \omega_i^*) = \arg \min_{\substack{a_i^* \in \mathcal{U}_a \\ \omega_i^* \in \mathcal{U}_\omega}} \frac{1}{2} \left(\|a_i^* - a_i\|^2 + \|\omega_i^* - \omega_i\|^2 \right) \quad (40)$$

s.t. (38) and (39).

Note that the CBF-QP controller above is designed in a minimal invasive manner, building upon the nominal controller. The CBFs modify the system behavior when the action specified by the nominal controller conflicts with the safety constraints, in which case a collision is imminent.

Altogether, the over safe platooning control strategy is outlined in Algorithm 2.

Algorithm 2: Path planning and platooning control

- 1 **Initialize:** vehicle state, estimator state, safety margin ε , safety distance D_{safe} , input constraints, and control parameters $k_1, k_2, c_0, c_1, c_2, \gamma_1, \gamma_2, \mu_1$, and μ_2 .
 - 2 Update the safe path $C'_0(\phi, \varrho)$. // Execute Algorithm 1
 - 3 Compute the control command ω_0 and a_0 for the leader vehicle by (31).
 - 4 Update the estimator state $\hat{\xi}_i$ and $\hat{\eta}_i$ by (32).
 - 5 Compute the nominal controller ω_i and a_i for the follower vehicles by (34) and (35).
 - 6 Update the safety constraints by (37) and (39).
 - 7 Compute the safe platooning controller (a_i^*, ω_i^*) by solving the QP in (40).
 - 8 Update the vehicle state using the dynamics in (1).
-

Remark 9: Model predictive control (MPC) has been widely used for vehicle platooning purposes, offering guaranteed safety and optimal performance. However, it should be mentioned that MPC methods often come with inherent computational costs in optimization over a receding horizon. In contrast, the proposed CBF-based QP control approach optimizes over a single step and well-suited for systems with nonlinear affine dynamics, where the resulting QPs are typically convex and computationally efficient to solve. Furthermore, the minimal intervention principle ensures that optimization is invoked only when the nominal controller violates the safety constraints. More importantly, it can handle unknown time-varying flowfields by integrating robust adaptive control design into the nominal component, thereby enhancing adaptability in uncertain environments.

V. NUMERICAL EXPERIMENTS

In this section, a platoon consisting of four follower vehicles, indexed by $i = 1, \dots, 4$, and one virtual leader, indexed by 0, will be considered for numerical experiments.

The platoon is tasked with cooperative and automated driving along a nearly 9 km curved path, while maintaining a desired spacing $d_0 = 10$ m. A superellipse curve is selected as an alternative path for a real-world highway shown in Fig. 7, which includes both bends and straight lines and is constructed as $C_0(\phi) := (x/a)^6 + (y/b)^6 = 1$, $\phi \in [0, \pi)$, with $a = 3$ km and $b = 2$ km. The acceleration and angular velocity bounds are set as $a_{\min} = -3$ m/s², $a_{\max} = 5$ m/s², and $\omega_{\min} = -\omega_{\max} = -0.2$ rad/s.

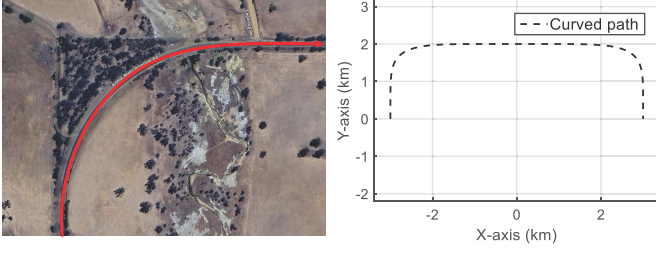


Fig. 7. The curved path employed for numerical experiments.

A. Performance of the Proposed Path Planning Algorithm

We first evaluate the proposed path planning algorithm in the path-following environment with two static large obstacles on the desired (original) path C_0 . The diamond obstacle with width 600 m is initialized at $p_{ob,1} = [0, 2000]$ m; and the circular obstacle is initialized at $p_{ob,2} = [1506, 1995]$, with radius $R_{ob,1} = 500$ m, as shown in Fig. 5. The tubular safe margin is set to $\varepsilon = 0.15$. The safe path associated to the platoon is generated by scaling the original path, with the optimal amplitude parameters $A_{q,1} = 0.5205$ and $A_{q,2} = 0.3384$. Fig. 5 illustrates the safe path in this case, showing that both the safe path and its corresponding safe region circumvent the obstacles.

In practical applications, GPS measurement noises are unavoidable, leading to inaccuracies in obstacle detection. To assess the robustness of the proposed path planning algorithm under such conditions, we further conduct Monte Carlo simulations with random measurement noises of a zero-mean normal distribution and covariance $\mathbf{R}_p = \text{diag}\{5, 5\}$. A total of 100 Monte Carlo runs are performed and the results are presented in Fig. 8. Thanks to the robust design of the safe region, the planned paths can effectively avoid obstacles, demonstrating the effectiveness and robustness of the proposed path planning algorithm in handling measurement noises.

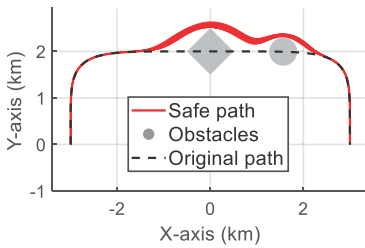


Fig. 8. Safe paths generated under 100 Monte Carlo simulations.

B. Performance of the Proposed CBF-QP Controller

We next assess the efficacy of the proposed CBF-QP controller in managing the CAV platoon. The follower vehicles are initialized off the designated path, and subject to unknown time-varying flowfields parameterized by (11), which are specified as follows: $f_{a_i} = 100\lambda_i^2 \tan \frac{\beta_i}{2}$, $f_{b_i} = \text{diag}\{\cos(0.5 \cdot i \cdot x_i), \sin(0.5 \cdot i \cdot y_i)\}$, and $\vartheta_i(t) = [\sin(0.1 \cdot i \cdot t), \cos(0.1 \cdot i \cdot t)]^T$. The safety distance for each vehicle pair is set as $D_{\text{Safe}} = 3$ m. The control parameters for the nominal controller are selected as $k_1 = k_2 = 5$, $c_0 = 1$, $c_1 = 80$, $c_2 = 10$, $\gamma_1 = \gamma_2 = 5$, and $\mu_1 = \mu_2 = 20$.

Additionally, for uniform speed driving, the reference signal is produced by $\dot{\eta}_* = \tau_* = 0$.

To further demonstrate the merit of the proposed controller, two typical V2V communication topologies are considered: the predecessor following (PF) communication topology (where each follower vehicle receives information only from the vehicle directly ahead) and the leader predecessor following (LPF) communication topology (where each follower vehicle receives information from both the vehicle directly ahead and the platoon leader). It is readily verified that both PF and LPF topologies satisfy Assumption 1. An additional square moving obstacle with side length 200 m is initialized at $p_{ob,3} = [-400, 1400]$ m, where the corresponding parameter $A_{q,3}$ is calculated online by Algorithm 1.

Implementing the proposed CBF-QP controller (40) together with the path planning algorithm, the corresponding results for the PF and LPF topologies are presented in Fig. 9 and 10, respectively. It can be observed that 1) the platoon is efficiently and safely maneuvered along the desired path, successfully avoiding the three moving/static obstacles simulated while maintaining the desired gap, as shown in Figs. 9a and 10a; and 2) the path-following errors λ_i on each CAV remain within the desired bound ε over time, as illustrated in Figs. 9b and 10b. Additionally, Figs. 9e and 10e demonstrate that the inter-vehicle spacing $\xi_{i-1} - \xi_i$ consistently exceeds the safety threshold $D_{\text{Safe}} = 3$ m, confirming the safety requirements (15) and (16). On the other hand, significant fluctuations are observed from the subplots (d), (e), and (f) of Figs. 9 and 10 between $t = 40$ s and $t = 60$ s. These fluctuations arise from changes in path curvature as the path planning algorithm scales and deforms the path to avoid obstacles. Furthermore, we evaluate the platooning performance under the PF and LPF topologies by comparing the root mean squared error (RMSE) and mean absolute error (MAE). As shown in Table I, it is evident that the platoon responses under the LPF topology generally outperform those under the PF topology, as it consistently achieves lower error values across all performance indices. This improvement is primarily due to the LPF topology's ability to relay the leader's motion information promptly to each follower. Finally, Figs. 9g and 10g indicate that the control inputs remain bounded as (2).

TABLE I
QUANTITATIVE COMPARISON OF PF VS. LPF TOPOLOGY

Performance Index	PF	LPF
RMSE (spacing error $\xi_0 - \xi_i - i \cdot d_0$)	9.2384	7.9234
RMSE (speed tracking error $\eta_0 - \eta_i$)	3.2257	2.8361
MAE (spacing error $\xi_0 - \xi_i - i \cdot d_0$)	3.0586	2.4824
MAE (speed tracking error $\eta_0 - \eta_i$)	1.6632	1.5018

C. Ablation and Comparative Studies

An ablation study on the effect of the CBFs for both PF and LPF topologies is conducted. The time histories of the inter-vehicle spacing are depicted in Fig. 11. As shown, disabling the CBF module leads to the violation of the safety constraints, particularly during obstacle avoidance maneuvers.

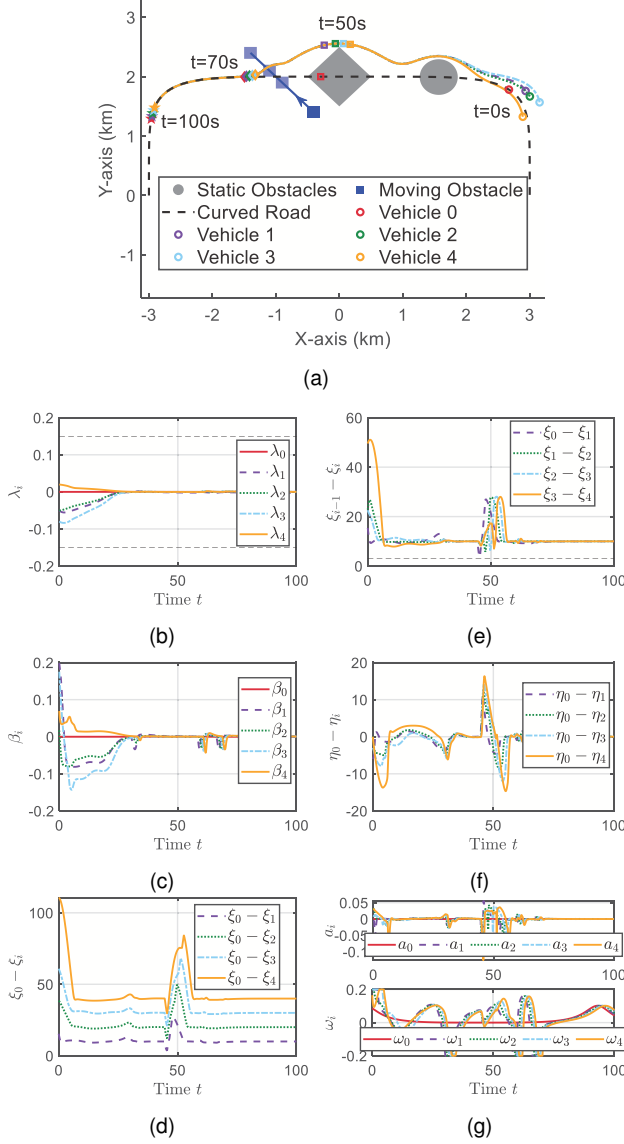


Fig. 9. Platoon responses under the proposed CBF-QP controller (40) over the *PF* topology: (a) Trajectories of the platoon at different times with obstacle avoidance, where $\circ$, $\square$, $\diamond$, and $\star$ denote the vehicles' positions at $t = 0s$, $t = 50s$, $t = 70s$, and $t = 100s$, respectively; (b) Path-following errors λ_i on each follower vehicle i ; (c) Angle errors β_i on each follower vehicle i ; (d) Relative spacing $\xi_0 - \xi_i$ between each follower vehicle i and the leader 0; (e) Spacing $\xi_{i-1} - \xi_i$ between every two consecutive vehicles; (f) Speed tracking errors $\eta_0 - \eta_i$ on each follower vehicle i ; (g) Acceleration and angular velocity inputs a_i and ω_i on each follower vehicle i .

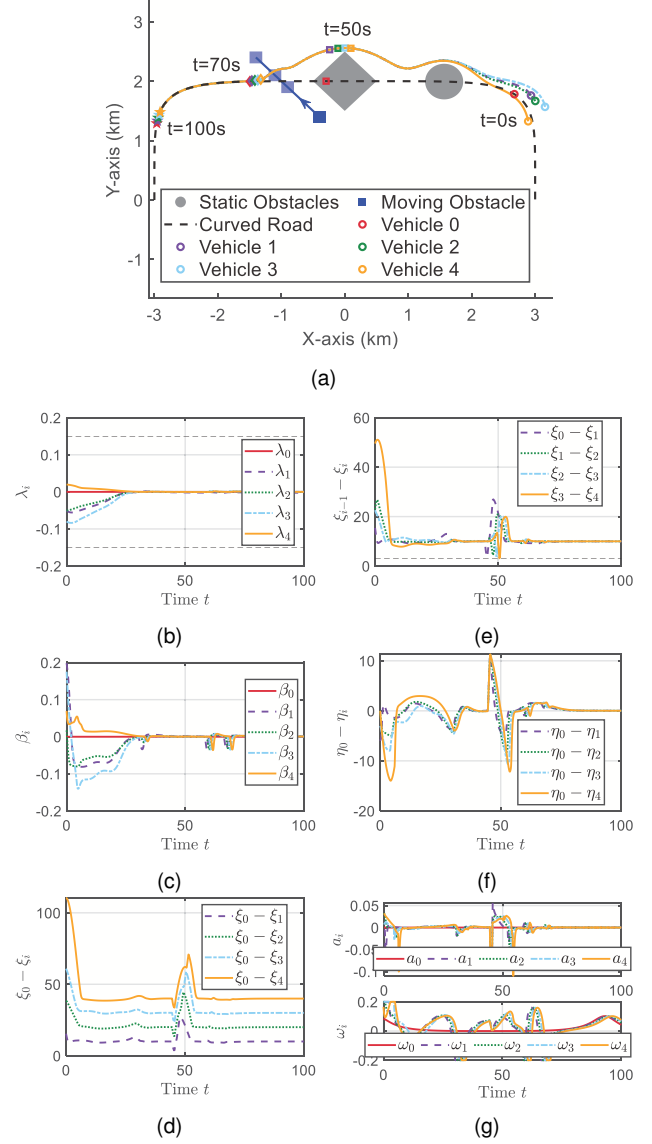


Fig. 10. Platoon responses under the proposed CBF-QP controller (40) over the *LPF* topology: (a) Trajectories of the platoon at different times with obstacle avoidance, where $\circ$, $\square$, $\diamond$, and $\star$ denote the vehicles' positions at $t = 0s$, $t = 50s$, $t = 70s$, and $t = 100s$, respectively; (b) Path-following errors λ_i on each follower vehicle i ; (c) Angle errors β_i on each follower vehicle i ; (d) Relative spacing $\xi_0 - \xi_i$ between each follower vehicle i and the leader 0; (e) Spacing $\xi_{i-1} - \xi_i$ between every two consecutive vehicles; (f) Speed tracking errors $\eta_0 - \eta_i$ on each follower vehicle i ; (g) Acceleration and angular velocity inputs a_i and ω_i on each follower vehicle i .

This demonstrates the critical role played by the CBF in maintaining inter-vehicle safety and enforcing collision avoidance.

The proposed CBF-QP control approach is further compared with a benchmarking model predictive control (MPC) algorithm (with a horizon of 5) under the same setup. As presented in Table II, the CBF-QP method incurs significantly lower computational cost due to its single-step optimization structure. In terms of control performance, the CBF-QP method achieves better results in longitudinal aspect, attributed to its robust adaptive component which effectively mitigates the effects of unknown time-varying flowfields.

D. Extension to A Mixed Traffic Scenario

We finally demonstrate the portability and robustness of the proposed platooning control scheme under a single-lane mixed traffic scenario. Specifically, we introduce a non-convex superellipse segment characterized by $C_0(\phi) := \alpha (2xy)^n + \beta (x^2 - y^2)^n = 1$, $\phi \in [0, \frac{3\pi}{2})$, where the shape parameters are set to $\alpha = \beta = 2$ km and $n = 8$. Unlike the simple bend-straight-bend curve used in the baseline experiments, this geometry exhibits substantial curvature variations. In addition, we incorporate a vehicle traveling ahead of the platoon in the same direction and lane at a constant speed of 60 km/h, which is lower than the nominal cruising speed of the platoon. The road boundaries are placed symmetrically at ± 3.5 m from the

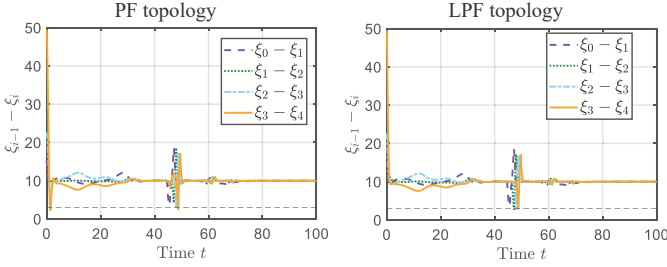


Fig. 11. Ablation study on the effect of the CBFs.

TABLE II
COMPARATIVE RESULTS

Performance Index	CBF-QP	MPC
Computational cost (each step) ¹	8ms	32ms
Path following error λ_i	7.4218e-04	0.0283
Heading angle error β_i	0.0026	0.0078
Spacing error $\xi_0 - \xi_i - i \cdot d_0$	0.2036	0.2104
Speed tracking error $\eta_0 - \eta_i$	0.4074	0.4091

¹On Matlab 2024b (with solver 'fmincon') on an Intel(R) Core(TM) i5-13400F CPU @2.50GHz.

lane centerline. All other settings remain unchanged.

Implementing the proposed Algorithm 2, the real-time movements of each CAV are shown in Fig. 12. The snapshots evidently illustrate that, despite the continuously varying road curvature, the platoon maintains the desired inter-vehicle spacing, safely overtakes the slow-moving vehicle ahead in the same lane, and consistently stays within the road boundaries.

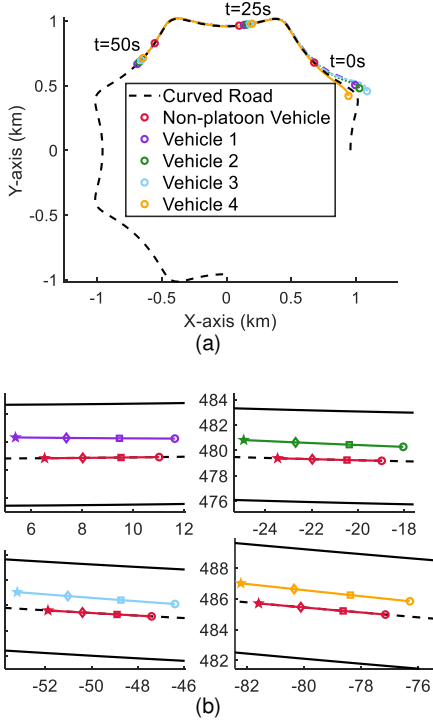


Fig. 12. Platoon responses in the mixed traffic scenario: (a) Platoon movement; (b) Snapshots, where $\circ$, $\square$, $\diamond$, and $\star$ mark the vehicles' positions at successive time instants, and the black solid lines depict the road boundaries.

VI. CONCLUSION

The combined longitudinal and lateral control problem for CAV platooning with safety requirements has been addressed.

Each vehicle dynamics have been described by a nonholonomic model subject to time-varying parametric flowfields. In order to achieve obstacle avoidance, a scaling-based path planning algorithm have been first developed to generate safe paths for the platoon vehicles. Then, a nominal controller has been designed such that the combined longitudinal and lateral control objectives are fulfilled. Furthermore, CBFs have been constructed to ensure safety requirements, and the control inputs have been determined by the QP solvers in a minimal invasive way. Finally, numerical experiments have demonstrated the efficacy of the proposed control design.

Future research will focus on incorporating dynamic traffic scenarios such as merging, splitting, and lane changing, where CAVs must negotiate priorities in real time while maintaining safety and efficiency. As another potential extension, it would be significant to explore the impacts of intermittent sampling and communication [47] and spatiotemporal disturbances [48] on the stability and performance of the proposed framework. This will enhance the robustness and practical applicability of the system in real-world CAV networks.

APPENDIX A: PROOF OF THEOREM 1

Consider the Lyapunov function $V = \frac{c_1}{2} \sum_{i=1}^N \lambda_i^2 - \sum_{i=1}^N \ln(\cos^2 \frac{\beta_i}{2}) + \frac{c_2}{2} \sum_{i=1}^N (\xi_i - \hat{\xi}_i - d_i)^2 + \frac{1}{2} \sum_{i=1}^N (\eta_i - \hat{\eta}_i)^2 + \frac{1}{2\gamma_1} \sum_{i=1}^N (\ell_{\vartheta_i} - \hat{\vartheta}_i)^T (\ell_{\vartheta_i} - \hat{\vartheta}_i) + \frac{1}{2\gamma_2} \sum_{i=1}^N (\delta_{\vartheta_i} - \hat{\delta}_i)^2$, where $\gamma_1, \gamma_2 > 0$ are the adaptation gains, c_1, c_2 are positive scalars, and $d_i = i \cdot d_0$. The time derivative of V along (26), (27), (29), and (30) is

$$\begin{aligned} \dot{V} = & \sum_{i=1}^N \tan \frac{\beta_i}{2} \left(\Delta_{i,1} - f_{a_i} f_{b_i} \hat{\vartheta}_i - f_{a_i} f_{b_i} \Delta_{\vartheta_i}(t) - \omega_i \right) \\ & + \sum_{i=1}^N (\eta_i - \hat{\eta}_i) \left(\frac{\partial \xi_i}{\partial s_i} a_i + \Delta_{i,2} \right) + \sum_{i=1}^N (\ell_{\vartheta_i} - \hat{\vartheta}_i)^T \\ & \times \left(-\frac{1}{\gamma_1} \dot{\vartheta}_i - \tan \frac{\beta_i}{2} f_{a_i} f_{b_i} \right) - \frac{1}{\gamma_2} \sum_{i=1}^N (\delta_{\vartheta_i} - \hat{\delta}_i) \dot{\delta}_i \end{aligned}$$

with $\Delta_{\vartheta_i}(t) = \vartheta_i(t) - \ell_{\vartheta_i}$ being the varying perturbation, and

$$\begin{aligned} \Delta_{i,1} = & -2c_1 \lambda_i \|\nabla \lambda_i\| v_i \cos^2 \frac{\beta_i}{2} - v_i (c_{a,i} \cos \beta_i + c_{b,i} \sin \beta_i) \\ & - c_2 (\xi_i - \hat{\xi}_i - d_i) \eta_i \left(\sin \beta_i + 2 \frac{\partial s_i}{\partial \lambda_i} \|\nabla \lambda_i\| \cos^2 \frac{\beta_i}{2} \right) \\ \Delta_{i,2} = & \frac{\partial^2 \xi_i}{\partial s_i^2} v_i^2 \left(\cos \beta_i - \frac{\partial s_i}{\partial \lambda_i} \|\nabla \lambda_i\| \sin \beta_i \right) - \dot{\eta}_i \\ & + c_2 (\xi_i - \hat{\xi}_i - d_i). \end{aligned} \quad (41)$$

According to Assumption 3 and noting that $|\Delta_{\vartheta_i}(t)| \leq \delta_{\vartheta}$, we can obtain from $\tan \frac{\beta_i}{2} f_{a_i} \geq 0$ that $\tan \frac{\beta_i}{2} f_{a_i} f_{b_i} \Delta_{\vartheta_i}(t) \leq c_0 \tan \frac{\beta_i}{2} f_{a_i} \delta_{\vartheta}$. Then, it follows that

$$\begin{aligned} \dot{V} \leq & \sum_{i=1}^N \tan \frac{\beta_i}{2} \left(\Delta_{i,1} - f_{a_i} f_{b_i} \hat{\vartheta}_i + c_0 f_{a_i} \hat{\delta}_i - \omega_i \right) \\ & + \sum_{i=1}^N (\eta_i - \hat{\eta}_i) \left(\frac{\partial \xi_i}{\partial s_i} a_i + \Delta_{i,2} \right) \end{aligned}$$

$$+ \sum_{i=1}^N (\ell_{\vartheta_i} - \hat{\vartheta}_i)^T \left(-\frac{1}{\gamma_1} \dot{\hat{\vartheta}}_i - \tan \frac{\beta_i}{2} f_{a_i} f_{b_i} \right) + \sum_{i=1}^N (\delta_{\vartheta_i} - \hat{\delta}_i) \left(-\frac{1}{\gamma_2} \dot{\hat{\delta}}_i + c_0 \tan \frac{\beta_i}{2} f_{a_i} \right). \quad (42)$$

Design the updating laws of $\hat{\vartheta}_i$ and $\hat{\delta}_i$ as

$$\dot{\hat{\vartheta}}_i = -\gamma_1 \tan \frac{\beta_i}{2} f_{a_i} f_{b_i}, \quad \dot{\hat{\delta}}_i = \gamma_2 c_0 \tan \frac{\beta_i}{2} f_{a_i}. \quad (43)$$

Substituting (34), (35), and (43) into (42), one can obtain that

$$\dot{V} \leq -k_1 \sum_{i=1}^N \frac{\sin^2 \frac{\beta_i}{2}}{\cos \frac{\beta_i}{2}} - k_2 \sum_{i=1}^N (\eta_i - \hat{\eta}_i)^2 \leq 0. \quad (44)$$

The set $\Phi_1 = \{(\lambda_i, \beta_i, \xi_i - \hat{\xi}_i - d_i, \eta_i - \hat{\eta}_i, \ell_{\vartheta_i} - \hat{\vartheta}_i, \delta_{\vartheta_i} - \hat{\delta}_i) \mid V \leq s_0\}$ is closed by continuity, for some $s_0 > 0$. Note that $|\lambda_i| \leq \sqrt{2s_0/c_1}$, $|\beta_i| \in (-\pi, \pi)$, $|\xi_i - \hat{\xi}_i - d_i| \leq \sqrt{2s_0/c_2}$, $|\eta_i - \hat{\eta}_i| \leq \sqrt{2s_0}$, $|\ell_{\vartheta_i} - \hat{\vartheta}_i| \leq \sqrt{2s_0\gamma_1}$, and $|\delta_{\vartheta_i} - \hat{\delta}_i| \leq \sqrt{2s_0\gamma_2}$ indicate that the set Φ_1 is compact. Therefore, the closed-loop system has a unique solution on the set Φ_1 . In this case, the closed-loop error system is

$$\begin{aligned} \dot{\lambda}_i &= -\|\nabla \lambda_i\| v_i \sin \beta_i \\ \dot{\beta}_i &= -k_1 \sin \frac{\beta_i}{2} - f_{a_i} f_{b_i} (\vartheta_i - \hat{\vartheta}_i) - c_0 f_{a_i} \hat{\delta}_i \\ &\quad + 2c_1 \lambda_i \|\nabla \lambda_i\| v_i \cos^2 \frac{\beta_i}{2} + c_2 (\xi_i - \hat{\xi}_i - d_i) \\ &\quad \times \eta_i \left(\sin \beta_i + 2 \frac{\partial s_i}{\partial \lambda_i} \|\nabla \lambda_i\| \cos^2 \frac{\beta_i}{2} \right) \\ \dot{\xi}_i - \hat{\xi}_i &= \eta_i \left(\cos \beta_i - \frac{\partial s_i}{\partial \lambda_i} \|\nabla \lambda_i\| \sin \beta_i \right) - \hat{\eta}_i \\ \dot{\eta}_i - \hat{\eta}_i &= -k_2 (\eta_i - \hat{\eta}_i) - c_2 (\xi_i - \hat{\xi}_i - d_i) - \tau_* \\ &\quad + \mu_1 \sum_{j \in \mathcal{N}_i} a_{ij} (\hat{\xi}_i - \hat{\xi}_j) + \mu_2 \sum_{j \in \mathcal{N}_i} a_{ij} (\hat{\eta}_i - \hat{\eta}_j). \end{aligned} \quad (45)$$

Since the controller (34) and (35) yields that V is non-increasing along the trajectories of the system, we conclude that if the initial value of V is finite, the entire solution of V stays in Φ_1 , which implies that $\lim_{t \rightarrow \infty} V = 0$. Consequently, the trajectory of (45) will converge to the region $E_1 = \{(\lambda_i, \beta_i, \xi_i - \hat{\xi}_i - d_i, \eta_i - \hat{\eta}_i, \ell_{\vartheta_i} - \hat{\vartheta}_i, \delta_{\vartheta_i} - \hat{\delta}_i) \mid V = 0\}$, then we have $\lambda_i = 0$, $\beta_i = 0$, $\xi_i - \hat{\xi}_i = 0$, and $\eta_i - \hat{\eta}_i = 0$ on E_1 . Recalling the exponentially convergent property of $\hat{\xi}_i - \xi_0$ and $\hat{\eta}_i - \eta_0$ with the help of the distributed estimator (32), we conclude that the control objectives (12)–(14) are achieved. This completes the proof. $\square$

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